Engineering & Ethics Analysis

Violated Engineering Principles:

- **Fail-Safe Design:** Systems should fail into a safe state. ETCS-i failed into a wide-open throttle state.
- **Separation of Concerns:** 10,000 global variables meant different vehicle subsystems could corrupt each other's memory space.
- **Simplicity (KISS):** The cyclomatic complexity of the throttle control modules was untestable.

Actor Responsibility Map:

- **Actor:** Toyota Management / Executives
  - **Responsibility:** Duty to protect public safety and comply with regulators.
  - **Missed Action:** Did not disclose "sticky pedal" data; blamed drivers instead of investigating anomalies.
  - **Consequence:** Preventable fatalities, $1.2B DOJ fine.
- **Actor:** Software Engineering Team
  - **Responsibility:** Design verifiable, modular, and safe code.
  - **Missed Action:** Failed to follow coding standards (e.g., MISRA C); used unbounded recursion and massive global variables.
  - **Consequence:** Unpredictable real-time operating system (RTOS) behavior leading to task death.

Ethics (C10): Breached IEEE/ACM Code of Ethics regarding prioritizing public safety, disclosing system limitations, and refusing to sign off on unsafe architectures. Engineers had a duty to escalate the lack of a hardware brake override.